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Method and system for generating dynamic user experience applications

Abstract

Systems and methods may be provided for generating applications that may be agile, personalized, quickly delivered, and capable of being seamlessly integrated across an organization. The behavior and functionality of the applications (e.g., user interfaces therein) may be tailored specifically to individual users in response to learned user preferences. Consequently, these dynamic user experience (UX) applications may be rapidly deployed and capable of providing a satisfactory yet complete user experience across one or more applications. The methods and systems may include receiving a user objective, selecting a path associated with one or more steps, generating a dynamic UX application based on the steps, transmitting the dynamic UX application to the user, and displaying the dynamic UX application.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of, and claims priority to U.S. patent application Ser. No. 17/750,554, filed on May 23, 2022, now U.S. Pat. No. 11,870,875, issued Jan. 9, 2024, which is a continuation of, and claims priority to U.S. patent application Ser. No. 16/001,553, filed on Jun. 6, 2018, now U.S. Pat. No. 11,340,872, issued May 24, 2022, which is a nonprovisional of, and claims priority to, U.S. Provisional Application No. 62/535,613, filed Jul. 21, 2017, the entire contents of which are incorporated herein by reference.

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to techniques for providing computer-implemented applications, and more specifically, to techniques for generating and executing dynamic user experience within applications.

BACKGROUND

(2) Typically, an organization may offer a number of computer implemented applications (e.g., web applications, mobile applications, and/or desktop software applications) via which users (e.g., the organization's customers, clients, and/or employees) may interact with the organization and its offered services and/or products. For example, the organization may offer an application via which a user may apply for a credit card, mortgage, or another financial product. The application may include one or more user interfaces (UI) via which the user may provide personal information relevant to acquiring the product.

(3) With regard to applications, a need exists for the organization to provide a satisfactory and consistent user experience (UX) for the users within one application, and/or across multiple applications offered by the organization. In this context, "user experience" may generally refer to the systems and practices via which one or more users may interact with a system or product (e.g.,

an application), and the UX may include one or more user interfaces (UI) via which the application may present information to users and/or receive user input (e.g., information or commands).

However, the methods and systems of generating applications (and consequently, the applications themselves) may be affected by a number of challenges that may negatively affect UX with regard to one or more applications. These challenges may include, but are not limited to, lack of agility, lack of personalization, slow delivery, and/or integration difficulty.

(4) First, application UX generally lacks business agility due to tight coupling in business process flows. Specifically, business logic may be contained in presentation and/or service layers (e.g., in both a client application and a server application). Such coupling may require changes to multiple components and extensive testing when business logic changes. Core business logic may be packaged in the same deployable file as an application, requiring regression testing over an entire application when only small changes are made to business logic.

(5) Second, applications generally lack personalized UX. For example, the same one or more user interfaces often provided to every user of an application, and the business logic that drives an interaction with a user may not be personalized or individualized, but rather may be similar or identical for all users. Changing the UX (e.g., changing a user interface) for one user may propagate unwanted changes to other users, and consequently, an individual or personalized UX may not be achieved.

(6) Third, current methods and systems for generating applications require too much time and cost to deliver. For instance, simple modifications to an application may cause time consuming, cascading changes across one or more applications. Consequently, an organization may be unable to quickly modify an application to improve UX in response to recognized market demands, opportunities, and/or usage analytics. Further, organization stakeholders may actively resist change due to the unforeseen consequences of modifying code. Even a “continuous delivery” solution (the process of small, incremental software releases), may not be possible due to the aforementioned tight coupling of UX and business process flows, and/or because the nature of the code modifications may require that code modifications be made “all or nothing” instead of continuously.

(7) Fourth, integration of multiple systems and/or applications within an organization may be difficult. Inconsistent application programming interfaces (APIs), data schemas, and aspects of product “look-and-feel” may emerge over time, particularly within large software enterprises that lack or do not closely follow standardization practices. Integration may also be particularly difficult in large software ecosystems comprising multiple sub-systems which have been developed independently over time and/or developed by different teams within a large and/or mature organization.

BRIEF SUMMARY

(8) In one aspect, a method for generating dynamic user experience applications includes receiving an objective of the user, selecting a path corresponding to the objective wherein the path is associated with one or more executable steps, generating a dynamic application by analyzing the one or more executable steps, transmitting at least a portion of the dynamic application to a device of the user, and displaying the at least the portion of the dynamic application in the device of the user.

(9) In another aspect a computing system including one or more processors and one or more memories storing instructions is provided. When the instructions are executed by the one or more processors, they cause the computing system to receive an objective of the user, and select a path corresponding to the objective, wherein the path being associated with one or more executable steps. The instructions may further cause the computing system to generate a dynamic application by analyzing the one or more steps, transmit at least a portion of the dynamic application to a device of the user, and display the at least the portion of the dynamic application in the device of the user.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) The figures described below depict various aspects of the system and methods disclosed herein. It should be understood that each figure depicts an embodiment of a particular aspect of the disclosed system and methods, and that each figure depicts an embodiment of a particular aspect of the disclosed system and methods, and that each of the figures is intended to accord with a possible embodiment thereof. Further, wherever possible, the following description refers to the reference numerals included in the following figures, in which features depicted in multiple figures are designated with consistent reference numerals.

(2) FIG. 1 depicts an environment for generating dynamic UX applications, according to an embodiment;

(3) FIG. 2 depicts an example flow diagram for executing dynamic UX applications, in accordance with some embodiments;

(4) FIG. 3 depicts an example data flow diagram for implementing dynamic UX applications, in accordance with some embodiments;

(5) FIG. 4 depicts an example method of generating dynamic UX applications, according to an embodiment;

(6) FIG. 5 depicts an example environment for implementing workflows and steps in a dynamic UX application, in accordance with some embodiments;

(7) FIG. 6 depicts an example data flow diagram for implementing dynamic UX applications as application programming interfaces, in accordance with some embodiments; and

(8) FIG. 7 depicts an example flow diagram for workflow execution, according to some embodiments.

DETAILED DESCRIPTION

I. Overview

(9) The embodiments described herein relate to, inter alia, generating computer-implemented dynamic UX applications that are agile, personalized, quickly delivered, and capable of being seamlessly integrated across an organization (e.g., a large business entity). More specifically, techniques may be used to generate one or more dynamic UX applications, the behavior and functionality of which an organization may dynamically modify as the application is installed or running in a client computing device, without requiring redeployment of the application to the client users. Techniques may be used to generate a dynamic UX application to carry out business processes organized into a workflow or path comprising a series of steps. The dynamic UX applications generated according to techniques described herein may include behavior and functionality specific to the user of the application, may be rapidly deployed using conventional or proprietary deployment techniques, and may be capable of communicating and exchanging data in a common format with other applications deployed across an organization. Business information and UX information may be entirely decoupled, and may be independently evaluated/executed, and scaled horizontally.

(10) In some embodiments, a client application may be provided to a mobile computing device (e.g., a laptop, server, smartphone, wearable device, etc.) of a user. The client application may allow the user to construct a dynamic user experience application. The client application may cause the mobile computing device to access a remote computing device (e.g., a server), and may transmit instructions or messages to the remote computing device which cause a path management application, or workflow manager application, along with associated steps, to be created, modified, and/or stored in the remote computing device. Herein, the terms “path” and “workflow” may be used interchangeably. The path management application may create and/or modify canonical resources on the remote computing device, and may cause the canonical resources to be associated

with the paths and/steps. The use of canonical resources and canonical data types may allow for data consistency and consistent development and maintenance of APIs over time. The steps may define information necessary to accomplish a business goal, and the paths and/or steps may be associated with rules or other modules which may determine how the steps are displayed in the mobile computing device.

(11) The workflows, or paths, may be portions of a business process that may require human interaction. For example, “collect user personal information” may be the business process corresponding to a step within a path. Human interaction may include collection of data, approval of information presented to the user, etc. A business process may comprise one or more workflows, and workflows may be reused and shared among business processes. The reusability of workflows and steps is a primary benefit of the methods and systems described herein. Workflows and steps may be reused across organizations that do not share common owners to build a consistent user experience. For example, a workflow may be used by two corporations A and B, wherein A and B have no common shareholders. Workflows may be deployed in applications, in production and/or development environments. A workflow, or path, may include state information and/or metadata. For example, each workflow may include a current step pointer, which corresponds to the step within the workflow that the user is in the process of completing. In an embodiment, each user may be associated with a list of step progresses, wherein the user's progress with respect to each step is included in the list of step progresses. A ratio of the number of steps that the user has completed to the total number of steps within a workflow may be computed by a workflow manager. A report may be obtained from the workflow manager specifying which steps are complete and the respective completion state of each one. Workflows may maintain step references, rules governing workflow version adoption, and workflow auditing requirements. Each step reference may refer to a step and upstream dependencies of a step. Each step reference may connect available step canonical properties to the business step parameters for a workflow. The information specified by the step reference may be used by a workflow manager to notify an experience manager to discover properties, and which properties should be discovered. Each step in the workflow may be associated with a step reference that represents the step. Workflow state may be captured in records known as interactions which may be associated with users. For example, state information with respect to a user's use of a credit card application workflow may be encapsulated into an interaction.

(12) Paths, or workflows, may be executable (e.g., by a client application and/or server application) and may comprise a series of executable steps. Executable may indicate that the step is capable of being translated by a protocol handler into a form that accepts one or more user inputs, or that the step may be expressed in human-readable format, or that instructions within the step may be executed to cause an output to be produced. Steps within a workflow may be ordered by the sequence in which they are presented to the user, and may repeat (e.g., in a workflow execution loop). Steps may be shared between workflows to allow for modularity and creation of new workflows from existing steps. Workflow steps may obtain the information necessary to execute their business rules from business capability services (e.g., from APIs which provide information) or other locations (e.g., from a user). Steps may contain, or cache, data internally that may be necessary for their respective execution. Steps may be externalized (i.e., run in an external rule engine). Steps may be executed in parallel for performance reasons, and may be executed in multi-pass, and/or recursively. The execution time of steps may be tracked and traced for profiling or other purposes. Steps may or may not be individually secured components; therefore, the decision of whether or not a user is authorized to access a step may be made by a component separate from a step. Workflows and steps may be executed based on user roles—for example, a workflow may be executed differently depending on whether the accessing user is an agent, customer, or a customer service representative.

(13) Steps may be reusable building blocks (e.g., “register a customer”). Generally workflows

correspond to business processes, (e.g., “credit card application”) and steps may encapsulate business rules which form part of the workflow (e.g., “collect applicant's home address”). A step may be an individual executable component of a path. Each step may maintain properties required for its execution and its current internal state. A step may be associated with a step type. Types of steps may include active state, address standardization, always complete, basic info, and others. In practice, steps may be implemented in any suitable manner (e.g., as Java classes deployed in a Jar file within an application). Steps may include or be associated with step parameters that may be linked to one or more canonical types, which may respectively correspond to the inputs accepted (and in some embodiments, required) by the steps.

(14) In some embodiments, steps may include prerequisites. For example, a step in which a user submits a credit card application may not be able to be completed by a user until one or more prerequisite steps are completed. This order completion requirement may be enforced by a workflow, a step, and/or a workflow runtime (e.g., a workflow manager). Similarly, some steps may be optional, in that a user may not be required to complete them to proceed to a next step. For example, a workflow may comprise an optional “cross sell” step in which the workflow presents a user with an offer to purchase a related product, but the user's decision to reject, or not accept, the offer may not affect the user's ability to progress to the next step. The optionality and prerequisite nature of steps may be defined in workflow metadata. In this way, a given step may be optional in one workflow, but not another. In yet another embodiment, an optional step may become a required step based on information entered by the user. For example, entering information about the user's real property holdings may be optional, unless and until the user declines to enter information about the user's savings accounts. The business logic for implementing optional steps that become required may be stored within the step, according to an embodiment.

(15) In yet another embodiment, a workflow may be used by an intake facility to digitize paper records. For example, information from a customer received via phone, agent office, mobile, and postal mail may be manually input into an appropriate workflow. The customer may then be sent a link to the workflow, and may resume working on an application with the information manually input pre-populated and available for the customer's inspection/verification. Manual input may also be used for the purposes of entering data into underwriting systems. In some embodiments, a user may switch between multiple input devices and/or protocols (e.g., from a cellular telephone to a voice-activated system) while remaining within the same workflow, and/or while interacting with the same step.

II. Example Environments for Generating and/or Executing Generic Dynamic UX Applications

(16) With respect to FIG. 1, an example environment **100** in which dynamic user experience applications may be generated is depicted, according to an embodiment. Environment **100** may include various client-side and server-side components; specifically, environment **100** may include a remote computing system **102** (equivalently “server **102**”, or, “server device **102**”) which may be coupled to a mobile computing device **104** (equivalently, “client **104**”, or, “client device **104**”) via a network **106**.

(17) Server **102** may be a single server or a group of multiple servers and multiple servers **102** may be deployed wherein each may be representative of a business API or service. Using multiple servers **102** may provide a number of benefits, including isolation between business APIs, isolation of production and testing environments, and greater availability and stability to each respective business API. Using multiple servers **102** may also provide the ability to independently tune and scale each business API, the opportunity to expose a more traditional API on a per-service basis, and may provide a business specific endpoint. Multiple servers **102** may provide tools and metrics that can be more easily aligned to business processes, and may provide each business API with the ability to be independently versioned giving consumers greater control over adoption and funding.

(18) Alternatively, or in addition, server device **102** may be a multi-tenant device, wherein multiple business APIs may be implemented in a platform-as-a-service (PaaS) model. Advantages of

implementing dynamic UX applications as PaaS may include more seamless sharing of resources between business APIs, quicker implementation of changes for multiple APIs (i.e., all instances may be changed simultaneously), and quick implementation of changes to shared resources between business APIs (e.g., business rules, common services, reusable workflow steps, etc.).

(19) Server device **102** may include a processor **108**, RAM **110**, application storage **120** (e.g., one or more persistent memories), and application **122**, which may include software instructions (e.g., a web stack, Android package, etc.). While referred to in the singular, processor **108** may include any suitable number of processors of one or more types (e.g., one or more central processing units (CPUs), graphics processing units (GPUs), etc.). Generally, processor **108** may be configured to execute software instructions stored in one or more memories (e.g., stored in a persistent memory such as a hard drive or solid state memory) of server device **102**. RAM **110** may comprise solid state and/or transient memory and may store application storage **120**, including instructions (e.g., corresponding to application **122**). When processor **108** executes instructions corresponding to application **122**, RAM **110** may, for example, temporarily store the instructions and data required for execution of application **122**. Processor **108** may load instructions stored in application storage **120** or any other location into RAM **110**. Software instructions, when executed by processor **108**, may generate, create, and/or execute one or more dynamic UX applications, which may be loaded into RAM **110** or application storage **120**.

(20) Generally, server device **102** generates dynamic UX applications which may be accessed/utilized by client device **104**, and processes user requests (also referred to herein as queries) generated by applications (e.g., application **168**) in response to the user's use of those dynamic UX applications. Portions of the dynamic UX applications generated by server device **102** may be displayed on client device **104** (e.g., by application **168** and display **152**).

(21) In some embodiments, application **122** may be part of a larger application or set of applications. Application **122** may provide code and data (e.g., a dynamic UX application) to client devices. Code and data may include a fully-formed dynamic UX application, or requests/queries used to construct, modify, and/or view a dynamic UX application or portions of a dynamic UX application displayed in a client device. For example, server device **102** may generate web pages (e.g., Hypertext Markup Language (HTML) instructions, JavaScript instructions, Java Server Pages (JSP) instructions, mobile device application, and/or any other type of instructions suitable for defining the content and presentation of the code and data), and/or may include instructions of a plug-in, extension, and/or stand-alone software component that may be downloaded by client device **104**.

(22) Generally, server device **102** may provide users accessing application **122** with one or more user interfaces which may enable the users to interact with application **122**, and which in turn, may modify data associated with server **102**. To that end, server device **102** may transmit user interface (UI) components (e.g., style sheets, executable code, HTML or JavaScript widgets, etc.) to clients, which the clients may execute, and with which the respective users of those clients may interact. Those clients may transmit data back to server device **102** in the form of requests/queries, which may include data payloads.

(23) Server **102** may also include data storage **124**, workflow module **126**, step module **130**, and experience module **132**. Server **102** may also include a communication interface **134** and a protocol handler **136**, which, respectively, cause information to be transmitted to and from devices (including server **102**) via network **106**; and perform operations (translation, formatting, processing, etc.) on that information. The precise arrangement of the foregoing components may differ from that depicted in FIG. 1. For example, in some embodiments, step module **130** may be contained within data storage **124**, or may reside in a separate server. In some embodiments, experience module **132** may be included in another server, in some embodiments, and an API may link experience module **132** to modules of server **102**.

(24) Data storage **124** may be composed of any suitable persistent or transient memory (e.g., a hard

drive, random-access memory unit, etc.), and may store any number of modules including, without limitation, those depicted in FIG. 1. Data storage may include an electronic database (e.g. a relational database, key-value store, flat file, etc.) in which electronic records representing paths and steps may be stored, and associated. For example, a path may be stored in a one-to-one, one-to-many, or many-to-many relationship with one or more steps. A step may be stored in a similar relationship with respect to one or more canonical types. A user may configure and modify the respective relationships between workflows, steps, and canonical types using the methods and systems described herein.

(25) Workflow module **126** may receive requests/queries, in response to which application **122** may take actions and issue responses. For example, workflow module **126** may advance the current step pointer from one step to another, for example, if workflow module **126** determines based on an analysis of the query that a step has been successfully completed. Workflow module **126** may communicate information to experience module **132**. Similarly, workflow module **126** may terminate a given workflow, add a step to a path, create a path, associate one or more canonical types with a step, evaluate a step, etc. based on a respective instruction from application **122**.

(26) Generally, workflow module **126** may contain instructions for managing workflows, or paths. Workflow module **126** may include workflow metadata, such as a name, creation date and time, and unique identifier associated with each workflow. Workflow metadata may also include a list of steps associated with each respective workflow. Workflow module **126** may be communicatively coupled to a data store (e.g., an electronic database) in which workflows and steps are stored. Generally, step module **130** contains steps and instructions for executing steps. Step module **130** may include step metadata, such as a name, creation date and time, and unique identifier associated with the each step. Step metadata may include a list of workflows associated with, or which include, each respective step. Step module **130** may include, for each step therein, a list of inputs, and associated canonical types associated with each respective input.

(27) Workflow module **126** may contain one or more workflows and may manage the execution of the workflows. Workflow module **126** may include a table of all workflows indexed by user or session, and their respective states. Workflow manager **126** may include a step with multiple resource types and steps may be executed as part of a workflow or individually. Steps may be associated with a user interface (e.g., a web form) and may log system activities (e.g., an audit log or information journal). Logging information may include debugging information (e.g., originating business context, etc.).

(28) In general, any component may log information. For example, a user may be presented with a form that is part of a step by a client application, and may enter information into, and submit, the form. Completion of the step may require the user to enter data that the client may transmit via the application. The step may process the form and, optionally, make an entry in an audit system for audit purposes. In some embodiments, a step may process its input in a client (e.g., client **104**). The same form may be re-used in case a user edits or verifies information displayed in a form. Then, the data the user entered previously may be sent to the step, which may pre-populate the form. However, in this case, the step may not be re-executed if the information has not changed (i.e., if the user does not change the information). Similarly, an audit entry may not be made if the step does not re-execute. Workflows and steps may both create audit records that may include a business context (e.g., the name of a referring agent). Workflow module **126** may determine the readiness of a step for execution in response to any new information workflow manager **126** receives.

(29) While executing/processing a particular step, workflow module **126** may launch an external application. Workflow manager **126** may return the user to the executing step when the external application closes. For example, an application may receive a step generated by workflow module **126** that launches a PDF document in a PDF reader. Other examples may include launching external applications for electronic signatures, supporting documents, and customer searches. After

reviewing the PDF document, the user may close the PDF reader and the application **168** may continue executing. In an embodiment, execution of the external program may not block execution of workflow module **126** (i.e., execution of external programs may be asynchronous with respect to the execution of workflow module **126**). In an embodiment, workflow module **126** may support an event-based architecture, and may provide real-time notification of events to users. In some embodiments, the workflow module **126** may prevent step information from being stored, either temporarily or permanently.

(30) Step module **130** may execute a series of steps in sequence for the performance of a task. For example, to complete a business workflow, step module **130** may execute steps 1-5 within a workflow, in sequential order. Execution of each step may include a response/request loop between a server (e.g., server **102**) and a client (e.g., client **104**), or a single computing device. The execution of each step may include one or more evaluations of user experience instructions located in experience module **132**.

(31) Experience module **132** may be a module that generates user experiences dynamically based on individualized UX rules defined by backend users, or generated according to an algorithm. For example, during the execution of a step by step module **130** or another module, a user of an application may be presented with a different experience based on information known about the user and/or the user's activity within the application.

(32) In some embodiments, workflow module **126** and step module **130** may execute workflow and steps without any user experience whatsoever, to provide a user with what is, in effect, a purely “vanilla” or unstyled, or naïve experience. In other embodiments, an experience module **132** may be used which includes information about the customer and may personalize the customer's digital interactions with application **122** in much the same way that humans personalize human interactions with the same customers. It should be understood that the functioning of workflow module **126** and step module **130** may be entirely unchanged in the latter embodiment, because experience module **132** is able to enrich steps and workflows completely independently of the functioning of workflow module **126** and step module **130**. That is, dynamic user experience information may be added by the use of experience module **132** without any changes to, or effect on, step module **130** and workflow module **126**. To provide personalized experiences to the user, experience module **132** may fit the experience to the individual customer, determining how and when to display content based on the user's preferences, or profile. Determining how and when to display content may be based on a runtime determination of information associated with the user or a session associated with the user. Such determinations may be made based on trained machine learning models, in some embodiments.

(33) Experience module **132** may create a user experience once, in the case of a standalone application **168** delivered to a user, or may do so on a continuous/on-demand basis for dynamic user experience applications in a server (e.g., application **122**) or client (e.g., application **168**). Experience module **132** may be implemented using a suitable application framework (e.g., Ember.js). A standalone application may be an application package file compatible with a mobile operating system such as Android or iOS.

(34) Experience module **132** may create UX flows and request information from the user, and may send messages in canonical request format to protocol handler **136**, which may in turn generate responses and send the responses to client **104**. Experience module **132** may receive an indication of information source which indicates whether the information was directly entered by the user, discovered by a company service/source, retrieved from an external source (e.g., a vendor), or calculated (e.g., created by a business step). Based on the information source, experience module **132** may adjust the UX displayed to the user accordingly by, for example, presenting the user with an indication that the information was automatically populated, and encouraging the user to double-check or confirm the information. Such indications may take any suitable form (e.g., a textual message, the addition of color, the reordering of information in a table, etc.).

(35) Experience module **132** may use information received from workflow module **126** to make UX flow decisions and to avoid problems (e.g., deadlocks). In an embodiment, experience module **132** may receive the number of steps that are remaining in workflow module **126**, which have not yet been executed, and an indication of what the steps are. Experience module **132** may then determine whether to delay asking the user for information. For example, in some embodiments, another component (e.g., workflow module **126**) may be unable to proceed without the information that experience module **132** wants to delay.

(36) Experience module **132** may translate requests/queries for further information from the user made by workflow module **126** into an experience designed by experience architects. In an embodiment, behaviors and UX events may be stored within an interaction, wherein the interaction holds information about actions the user takes within the UI which can be used by experience module **132** to make decisions about which experience to deliver to the user in a given circumstance. Any user accessing application **122** may be provided with a different, or personalized, UX based on captured UX events, and other information. For example, experience module **132** may query another module for information which may be used to create a UX. Experience module **132** may learn another module that a particular user prefers a dark background in application A. Experience module **132** may then change the UX delivered to the user (e.g., a web application, standalone application, etc.) in application B to include a dark background. Experience module **132** may be associated with a content delivery network (CDN), UX rules database, and with one or more system of interaction electronic databases. Workflow module **126** may present the user with a UX that is obtained, in whole or in part, from experience module **132**. For example, experience module **132** may include an intelligent agent which assists users in selecting auto insurance coverage. The selection of UI/UX components may be based on information experience module **132** may access regarding the workflow and user. The intelligent agent may prompt a first user to select coverage from a drop-down menu. The intelligent agent may prompt a second user to select coverage by manipulating a slider UI control. The choices of information may be the same or differ, but the presentation to the user may be personalized. The personalization (e.g., drop-down or slider) may depend on interactions that the first and second users have had previously with experience module **132**. For example, in the past, a user may have responded more favorably to one UI control than another. Such favorable responses may have been captured by an interaction module or experience module.

(37) In an embodiment, experience module **132** may include instructions that make UX decisions based on information needed by workflow module **126**. For example, if only one step remains in a path, experience module **132** may provide an indication to user that the workflow is “almost complete!” A color or other visual indication may be used to denote that the path is nearing completion. In some cases, experience module **132** may adjust components used in the UI based on the confidence level of information collected. For example, if a user provides a telephone number that contains letters, then experience module **132** may show a UI control accentuated to designate the input as possibly invalid or suspect. Experience module **132** may make UX decisions based on the number of steps that are available for execution. For example, experience module **132** may display a different UI arrangement for a workflow comprising 25 steps than one that only contains two steps. For example, in the later example, the UX may consist of a single page, whereas in the former example, two or more pages may be used.

(38) According to an embodiment, experience module **132** may use information collected during research to present an UX that is likely to cause users to complete workflows. Information collected during research may include, for example, user survey data, user facial analysis data, or other information regarding operation of one or more applications via one or more users. For example, research may indicate that a significant amount of potential customers halt their progress through an auto quote workflow when asked, early on in the workflow, for their social security number or other personal data. Experience module **132** may use this sort of finding to determine an

optimal time to present or ask for a social security data (e.g., via a particular user interface) while another module returns data that may be useful in completing the quote as far as possible before the social security data may be needed. Additionally or alternatively, experience module **132** may determine an optimal manner of presenting or requesting particular information (e.g., whether to request/receive social security data via a visual data entry field, or via audio input/output). Although some of the foregoing examples involve varying UI controls within UX (such as sliders and dropdowns), other aspects of UX may also be varied. For example, the language of messages to the user may be changed, including slang and affectation, depending on user feedback.

(39) Protocol handler **136** may receive requests/queries from clients. Protocol handler **136** may examine requests, determine if each request is associated with a user or session, and if so, whether the user/session corresponds to an existing user/session. Sessions may be stored in a database communicatively coupled to server **102**, or client **104**. Protocol handler **136** may determine a network protocol and sub-protocol associated with a request, and may route the request to the workflow module **126** or another module, which may cause workflow module **126** to begin executing a workflow at a position (e.g., at a step). The position may be specified by a request, or may be associated with a user/session. In general, requests may be received by protocol handler **136** and responses may be sent by protocol handler **136**. As used herein, the term “request” generally refers to an electronic request object that represents an exchange of information between a user and an application via the network **106**. The terms “user request” and “request” may be used interchangeably. Requests and responses may be transferred between server **102** and clients (e.g., client **104**) via a common data interchange protocol (e.g. via a representational state transfer (REST) API) or a proprietary data interchange protocol. While some of the examples herein refer specifically to requests and responses pertaining to, for example, credit card and auto insurance applications, it should be appreciated that the techniques described herein are applicable to many other types of applications. For example, a gaming application created for a mobile device may support its own form of user requests and responses.

(40) In some embodiments, requests are stateless, and in others, requests may have state. A request may be user-initiated, wherein the request is based on, or generated in response to, a direct or indirect user action. For example, a user of a user device (e.g., server **102** or client **104**) may perform a click event on a form submit button using a peripheral device, or may make a selection via a capacitive touch screen, or perform any other suitable action indicating an initiation of a request. Instructions executing in the user device may process the user selection or event and, in response, generate and transmit a request (e.g., via network **106**). It should be noted that in some embodiments, the user device and the remote computing system may refer to the same device (e.g., server **102**), and that in other embodiments, the user device may be a client device (e.g., client **102**) linked to the remote computing system via an intermediate wired and/or wireless network.

(41) A request may include an operation, wherein the operation may include creating, retrieving, updating, and/or deleting one or more paths and/or steps. For example, the user may initiate a “CREATE” operation with respect to a path. To do so, a user may be required to enter path information by, for example, typing or otherwise entering information into a form fieldset. The operation and path information may be included in the request, and may be evaluated by workflow module **126**. Executing, or evaluating, the request may include mapping the operation (e.g., UPDATE) to a corresponding method of one or more of a plurality of electronic database systems, and depending on the request type, executing the request may include providing the method with one or more evaluation parameters, which may correspond to information in the request (e.g., a path ID, step ID, user ID, etc.).

(42) Similarly, requests may include information entered by a user during the execution of a dynamic user experience application accessed by a user. A user may access application **168**, and by utilizing various functionality of application **168** (e.g., by selecting a user interface widget) may cause application **168** to issue requests and receive responses. For example, a user may enter

information into a form, and press a “submit” button, which may cause the information to be transmitted and/or processed. In this case, the request may not include information modifying, deleting, or creating a step or workflow, and may only contain the information that the user entered. (43) More than one protocol handler **136** may be configured, and protocol handler **136** may be integral to a client and/or server in environment **100**. As stated, protocol handler **136** may receive a user request in response to behavior and events of a user (e.g., a mouse click or another event). Further, code executing in a client may cause user requests to be sent to server **102**, and may receive responses, potentially unbeknownst to the user. Requests may be sent by client **104** synchronously or asynchronously with respect to the execution of code in client **104**. As noted, in addition to translating requests between clients and applications, protocol handler **136** may route requests to application **122** components, and may also provide other modules with information pertaining to the capabilities of client devices. If a request type corresponds to a workflow initiation or resumption, then protocol handler **136** may route the request to workflow module **126**. Upon receiving a request, workflow module **126** may determine whether to resume an existing interaction or begin a new one based on the request contents. Workflow module **126** may determine whether an interaction already exists by analyzing an indication and/or metadata from client device **104** (e.g., a UUID and/or an HTTP header). If an existing interaction is to be used, then workflow module **126** may retrieve the existing interaction. If a new interaction is to be used, then workflow module **126** may initiate a new interaction. It should be noted that a user may also decide to cancel or terminate a workflow. Also, a step may contain logic that causes a workflow to terminate if a certain condition is met, or if a particular input is provided to the step by a user (e.g., by a user of client **104**). In some embodiments requests and responses may be respectively handled and issued by client device **104**, either exclusively or in combination with server **102**.

(44) Server **102** may also include a display device **138** and an input device **140**. Input device(s) **140** may include components that are integral to server **102**, and/or exterior components that are communicatively coupled to server **102**, to enable server **102** to read/retrieve inputs from the user via input device **140**. For example, input device(s) **140** may include a mouse, a keyboard, a trackball device, a microphone, scanner, etc. Display device **138** may also be either integral or external to server **102**, and may use any suitable display technology (e.g., LED, OLED, LCD, etc.). In some embodiments, input device(s) **140** and display device **138** are integrated, such as in a touchscreen display. Generally, input device(s) **140** and display device **138** combine to enable a user to interact with user interfaces provided by server **102**, for example, user interfaces included in, or provided by, application **122**.

(45) Client **104** may be any suitable user device (PC, notebook, laptop, desktop, smart phone, or tablet) and may run any suitable operating system. Client device **104** may include one or more client user interfaces (UIs), which the user may access and interact with (e.g., via any suitable web browser). Client UIs may include software (e.g., a dynamic UX application UI including a UI web application and/or model, a UI control library, etc.) owned or controlled by the company or entity operating server **102**, and the client UIs may be served to mobile computing device **104** via network **106** from remote computing system **102** (e.g., over HTTP). The client UIs may be implemented as standalone applications (e.g., as Android application packages (APK)) or as executable web code (e.g., HTML, JavaScript, and CSS). In some embodiments, a client UI may be a hybrid of application package code and executable web code. A client such as client **104** may also be referred to as a “user device”.

(46) While FIG. 1 depicts only a single client device **104**, it is understood that multiple different client devices (of different entities and/or users), each similar to client device **104**, may be in remote communication with server **102**. Further, multiple client devices **104** of different types, which may correspond to different user cohorts, may access server **102**. For example, in some embodiments, server **102** may accept requests from clients which, respectively, include backend, or administrative, users. Such users may be employees or agents of the proprietor of the methods and

systems described herein, and may use client **104** and the services provided by server **102** for the purpose of defining and administering workflows/paths, steps, canonical types, and other operations (i.e., to create dynamic user experience applications for use by other users) as described above.

(47) In the embodiment of FIG. **1**, client device **104** includes an input device **150** and a display device **152**. Input device(s) **150** may include components that are integral to client device **104**, and/or exterior components that are communicatively coupled to client device **104**, to enable client device **104** to accept inputs from the user. For example, input device(s) **150** may include a mouse, a keyboard, a trackball device, a microphone, scanner, etc. Display device **152** may also be either integral or external to client device **104**, and may use any suitable display technology (e.g., LED, OLED, LCD, etc.). In some embodiments, input device(s) **150** and display **152** are integrated, such as in a touchscreen display. Generally, input device(s) **150** and display **152** combine to enable a user to interact with user interfaces provided by client device **104**, for example, client UIs.

(48) Client device **104** may also include a processor **160**, a random-access memory (RAM) **162**, a communication module **164**, and an application storage **166**. Application storage **166** may include application **168**. Client device **104** may also include a data storage **170** and a camera **172**. Processor **160** may include any suitable number of processors and/or processor types. Processor **160** may include one or more CPUs and one or more graphics processing units (GPUs), for example. Generally, processor **160** may be configured to execute software instructions stored in application storage **166** (e.g., client UIs in application **168**).

(49) Application storage **166** may include one or more persistent memories (e.g., a hard drive and/or solid state memory), and store a number of applications including, for example, a web browser application implementing application **168**, in an embodiment. Application **168** may be a web application, or an application of a different type, such as a standalone executable, distributed mobile application, etc. Data storage **170** may include one or more persistent memories, and generally stores data used by applications stored in application storage **166**. For example, data storage **170** may store local copies of steps or other code/data downloaded from server device **102**.

(50) Data stored in application storage **166**, application **168**, and data storage **170** may be loaded into RAM **162** when processor **160** executes instructions stored in application storage **166**. Instructions implementing application **168** may be downloaded to application storage **166**, and when processor **160** executes application **168**, for example, RAM **162** may temporarily store the instructions and data required for its execution while application **168** displays the dynamic UX application on display **152** or otherwise operates on the dynamic UX application. When the user of application **168** uses application **168** to access server device **102**, scripts, applications, and/or other instructions of server device **102** (e.g., instructions associated with workflow module **126** and step module **130**) may be stored as local copies in RAM **162**. Application **168** may load components and data from remote computing system **102**, such as user interface components, and application data, into RAM **162**, and may store such components and data locally (e.g., in data storage **170**). Data storage **170** may include persistent or transient storage devices, including magnetic and/or solid-state storage devices.

(51) Mobile computing device **104** may transmit user requests to remote computing system **102** via network **106**. Requests may include data and metadata indications (e.g., a universally unique identifier (UUID) or other identifier), and may be triggered by a user action (e.g., via a user interaction with input device **150**) or by the occurrence of a condition (e.g., code in application **168**). A request or query may include an indication of one or more workflows/paths, and or associated steps. A request may include a tree data structure, wherein each workflow, or path, is a node comprising a plurality of other nodes and/or leaves (i.e., leaf nodes). A leaf node may be a node with no children, or a node with no back links. A node may represent a path, a step, or a sub-step (e.g., a step that includes additional steps). The entire tree of nodes may be downloaded from server **102**, modified in client **104**, and then uploaded to server **102**. In some embodiments, a tree

may be a specific type of tree, such as a binary tree, red-black tree, etc. When a path including steps is evaluated, evaluation may proceed according to a modified breadth-first search or depth-first search algorithm, or another tree-traversal algorithm.

(52) In some embodiments, a client application (e.g., client **168**) may modify the state of paths and steps within the overall system. Further, client device **104** may decorate requests with an indication of the type of request. For example, the request may be designated as a business resource request, a user information request, a workflow information request, a behavior and UX event, etc.

Communication between client device **104** and server device **102** may be implemented using a secure transfer protocol such as HTTPS.

(53) In operation, a backend user may open application **168** in a web browser or as a standalone application (e.g., in a mobile device, laptop, etc.). The user may be presented with a list of paths and steps, which may be pre-loaded into the application **168**, or retrieved via communication module **164**, network **106**, and communication interface **134**. The list of paths and steps may be presented to the user in any suitable format, and may be retrieved by the user submitting a request to a remote computing server, such as server **102**. The request submitted by the user may include an indication of one or more paths, and the remote computing system may select a set of active paths and respective associated steps based on the indication of one or more paths. The indication of one or more paths may be a simple list of identifiers corresponding to a path, such as integers, or a more complex indication, such as a nested structure. The set of active steps may be transmitted to a user device, wherein the user device is a client device other than the remote computing server, or wherein the user device is the remote computing server. That is, the user may be using the remote computing server directly by input and display devices directly coupled to the remote computing server, such as via display device **138** and input device **140**, or indirectly, via a client device and network, such as mobile computing device **104** and network **106**.

(54) In some embodiments, paths may be stored in a tree wherein the nodes of the tree correspond to paths and the leaves of each node represent sub-paths, and wherein each path includes one or more steps that may themselves be arranged in a tree, according to some embodiments. Path and steps may be transmitted to the user device as described above in a nested data structure, such as a hash map or any other suitable data structure which allows a hierarchical arrangement of paths with respect to steps to be preserved, and processed/displayed by a client application, such as application **168**. Path and steps information may be presented to the user in display **152** and the user may be able to select a path or a step within a step. Selection of a path or step may cause execution of the path/step to begin.

(55) In an embodiment, a first set of paths, each associated with a respective first set of one or more steps may be received in a user device, wherein each is associated with a respective first set of one or more steps. The user may select one of the paths. In response to the user's selection of a path, the path may execute the steps associated with the path. In some embodiments, the user may select a path and may be redirected, by a workflow manager, to a step that is in the middle of the path; i.e., a step that is not the first step in the path. Such redirection may be the result of the user's having already completed one or more steps within the path, and the step that the user is redirected to may be the first uncompleted step in the path. Execution of a path or workflow may include recursive execution of all steps within that path, in addition to resolution of dependencies between steps.

(56) Steps may include step logic, or rules, which the user may enter as a group of parameters, operators, and operands. The step logic may be executed at the time that the step is executed, to validate, transform, or process inputs. A step may return one or more values to a code or application executing the step. A step and the step's one or more respective step operators and step operands may be associated together in an electronic database. Relationships between steps may be defined, such as the hierarchical ordering of steps within a path tree, in some embodiments, or the linear ordering of steps in some other embodiments. The user may define dependencies between steps, in that a step A must be presented to the user, and valid output obtained, before a step B may be

presented to the user. Such dependence may be represented using an appropriate data structure (e.g., a stack or an acyclic graph/tree), in an electronic database or in a memory such as data storage **124**.

(57) A step may have a default (e.g., vanilla) user experience with which to associate a path and/or one or more steps within the path. Available user experiences may be published by experience module **132**, and may specify both the look and feel, ordering and arrangement, and tone/content of information delivered to users via the execution of steps within a path. Multiple versions of paths and steps may exist, which may be executed at different times, based on conditional criteria (e.g., the identity of the user, the time of day, randomly, according to a schedule, etc.). In some embodiments, technical users may construct steps, and non-technical users may build paths. In some embodiments, paths and steps may include logic preventing information from one business unit from being accessible in another, due to regulatory and or legal/compliance issues.

(58) It should be appreciated that in some embodiments, the user interface components displayed to users are completely uncoupled from the paths and steps represented by the user interface components, in that a user never specifies whether a particular step will use, for example, a drop-down list in HTML to solicit a user selection. Rather, a step may require a user selection from a list through the administrative user interface application **122**, and another component (e.g., experience module **132**) may determine the graphical user interface component(s) that will be used to accomplish the collection of the needed information, at the time the step is executed. Not only may the controls be subject to runtime determination by experience module **132**, but also the ordering and frequency of information collection may be determined by experience module **132**.

(59) Network **106** may provide connectivity between server **102** and client **104**, in some embodiments. Network **106** may be a single communication network, or may include multiple communication networks of one or more types (e.g., one or more wired and/or wireless local area networks (LANs), and/or one or more wired and/or wireless wide area networks (WANs) such as the Internet). Server device **102** may be accessible by multiple networks, for example by a company intranet and the Internet. Customers may be granted access to a portion of server device **102** (e.g., a bank credit card application) via the internet, whereas business partners, agents, and employees of the proprietor of server device **102** may be granted broader, or narrower, access to portions of server device **102** (e.g., to a path editor or experience editor). Access to additional services may be similarly granted or restricted by the proprietor of server device **102** on any suitable basis.

(60) With regard to FIG. 2, an example flow diagram **200** for executing workflows within a dynamic UX application is depicted, according to an embodiment. Flow diagram **200** may include a user **202** who may correspond to a user of mobile computing device **104**, a workflow module **204** which may correspond to workflow module **126**, an experience manager **206** which may correspond to experience module **132**, a profile **208**, a series of workflows **210**, and a step analyzer **212** which may correspond to step module **130**. Experience manager **206** may include a list of steps **214** which may correspond to a workflow, and a list of user interfaces **216** each of which may correspond, respectively, to list of steps **214**. In an embodiment, step analyzer **212** may execute each step in list of steps **214**, and may decorate each step with dynamic user experience information during the execution (e.g., while generating HTML). In another embodiment, step analyzer **212** may receive an already-executed step from workflow module **204**, and may decorate the output of the step (e.g., by parsing HTML contained therein). In some embodiments, list of user interfaces **216** may be transmitted, one step at a time, to user **202** as each step is executed by step analyzer **212**. In other embodiments, list of user interfaces **216** may be transmitted to user **202** as a whole, after all of list of steps **214** has been executed.

(61) In operation, user **202** may open or access an application such as application **122** or application **168**. Such an application may receive or retrieve a list of workflows, such as workflows **210**. As in the depicted embodiment, workflows **210** may include a number of workflows 1, 2, . . .

P, wherein P may be any integer. Each of respective workflows 1-P may include any number of steps, which may or may not be identical as between workflows. For example, as depicted, workflow **1** may include steps 1, 2, . . . N; workflow **2** may include steps 1, 2, . . . M; and workflow P may include steps W, X, . . . Z. In this example, steps 1 and 2 may be the same step both used in workflows 1 and 2, and N and M may be any integers. Steps W, X, and Z may be steps unique to workflow P. User **202** may select workflow **2**, which may correspond to a particular activity, or business process, such as “apply for a new credit card”. Such a selection may be made via any suitable means (e.g., by activation of a link, or UI control) as discussed with regard to FIG. **1**. Selection may cause a request to be made, in some embodiments, from a client device to a server device. The request may be received in the server and may cause execution of a first step in a workflow to begin. Execution of a first step may be based on a session of the user. As noted above, a “first step” may be the first step in a sequence, or the first uncompleted step (e.g., a step at any sequential position in a path). For example, the request may include the ID of a workflow (e.g., **2**) which workflow module **204** may use to retrieve a workflow. The list of steps in the retrieved workflow may then be analyzed by workflow module **204** to determine the appropriate step for execution. The list of steps in the retrieved workflow may correspond to list of steps **214**, and may be transmitted to experience manager **206** by workflow module **204**. Profile **208** may include information pertaining to, or related to, a user. For example, demographic information about the user (e.g., age, location, occupation, etc.) may be included. Profile **208** may also include use pattern information. For example, information related to the user's use of a client or server included in the system (e.g., server **102** and **104**) may be included in profile **208**. Demographic and usage pattern information may be used by step analyzer **212** to determine and/or generate the user interfaces in list of user interfaces **216**. List of user interfaces **216** may compose or comprise a dynamic UX application, in some embodiments.

(62) Execution of list of steps **214** may be performed by step analyzer **212**. An internal pointer may be analyzed to determine which step to execute first, and/or the state of each step may be analyzed by step analyzer **212** to determine which steps are eligible for execution. As discussed above, in some embodiments, steps of a certain type may be executed last (e.g., steps which ask for highly personal information, such as social security numbers). In some embodiments, the timing of execution of steps which require highly personal information may be affected by the type of workflow (e.g., by being deferred). For example, in a home loan workflow, highly personal information may be requested first, whereas in an auto loan workflow, highly personal information may be requested last. The timing of execution may be determined by user experience rules associated with experience manager **206**, wherein such rules are executed by step analyzer **212**. Step analyzer **212** may determine information about the user by receiving or retrieving information from profile **208**. In an embodiment, information contained in a session associated with a user may be used to determine the current, next, and/or previous step.

(63) Turning to FIG. **3**, an exemplary data flow diagram environment **300** for generating dynamic UX applications, is depicted, in accordance with an embodiment. Environment **300** may include a user **302**, user device **304**, and remote computing system **306**, which may correspond, respectively, to user **202**, mobile computing device **104**, and remote computing system **102**. Environment **300** may include a workflow module **308** and experience module **310**, which may respectively correspond to workflow module **204** and workflow module **126**, and experience manager **206** and experience module **132**.

(64) In operation, user **302** may open or access an application in user device **304**. User **302** may have a particular business objective in mind while doing so. For example, user **302** may be seeking a loan. Business objective of user **302** may be transmitted from user device **304** remote computing system **306** via a request object, as described above. In some embodiments, user device **304** may be the same device as remote computing system **306**. Remote computing system **306** may transmit the received business objective to workflow module **308**, wherein one or more workflows

corresponding to the business objective may be selected. For example, a business objective may include a search term such as “loan” and workflow module **308** may return a list of workflows which match the search term (e.g., auto loan, home loan, personal loan, etc.). The list of workflows may be passed to experience module **310**, where they may be personalized in accordance with preferences of user **302**, as described above, to create a dynamic user experience. As described above, the workflows may be encapsulated in an application container, or served to user **304** via a web server (e.g., as Java server pages, HTML, etc.). Each dynamic user experience may be transmitted back to user **304** by way of workflow module **308** and/or remote computing system **306**. User **302** may interact with the user experience delivered by environment **300** in furtherance of the business objective.

(65) With regard to FIG. **4**, a method **400** of generating a dynamic UX application is depicted, according to an embodiment. Method **400** may include receiving a business objective (block **402**). The business objective may include a search phrase, such as “credit card”. Alternately, the business objective may be determined with respect to an existing product (e.g., “increase liability coverage”). The business objective may be selected from a pre-existing list that is received by a user device such as user device **304**, or may be user-directed (e.g., a search term or phrase). Method **400** may include selecting one or more workflows, or paths, corresponding to the business objective (block **404**). For example, a user whose selected business objective is “decrease monthly premium” may cause a plurality of related paths to be selected, wherein each path respectively consists of multiple steps designed to reduce the user's monthly premium in a plurality of respective categories. For example, “reduce auto premium”, “reduce home premium”, and “apply discounts” may be three respective workflows corresponding to a single “decrease monthly premium” business objective. It should be appreciated that many more examples are envisioned. Method **400** may include determining, or generating, an experience with respect to one or more of the selected workflows (block **406**). As described above, determining/generating an experience may include an experience module such as experience module **310**, experience manager **206**, or experience module **132** executing one or more UX rules, and may also (or alternately) include examination of a user's prior use patterns and/or expressed preferences. Determining/generating an experience may include analyzing the protocol of the user's device. For example, the provided UX may differ based on whether the user is utilizing, for example, short message service (SMS). Similarly, determining/generating an experience may include analyzing the device capabilities of the user device (e.g., the screen resolution, network connectivity, available memory, etc.). Method **400** may include transmitting the generated UX to the user, and/or the user's device (block **408**). In some embodiments, the entire method **400** may be implemented in a single user device, and as such, transmission may be performed via a local loopback networking device, or via shared memory. In other embodiments, transmission may be performed via a wired or wireless network, such as network **106**. It should be appreciated that in some embodiments, the dynamic UX application may be pre-compiled and delivered to the user as a standalone executable. In other cases, the dynamic UX application may execute in a runtime, and portions of the application may be transmitted to the user in a request/response cycle via a network. Depending on the particular embodiment used, transmission patterns may differ. Method **400** may include displaying the dynamic UX in the device of a user (block **410**). Displaying the device may, as noted above, be performed by painting an output display device, such as display device **138** or display device **152**. In some embodiments, “display” may connote portions of responses being played back via a sound card and audio speakers integral to, or communicatively coupled to, a computing device of a user. That is to say, the “display” of information may differ, depending on the protocol and hardware the user utilizes to access the dynamic UX application. Protocol handler **136** may have built-in facilities for translating user experiences generated by user experience modules into the appropriate display format. In some embodiments, displaying the dynamic UX may include delivering hypertext and other multimedia sources/instructions to a browser or application executing in the

data storage of a user device, such as application storage **166**, and such execution may be performed by a processor (e.g., processor **160**). In some embodiments, dynamic UX instructions may take the form of styling and behavior instructions implemented in, respectively, cascading style sheets (CSS) and JavaScript.

III. Example Environments for Validating and Completing Workflows and Steps

(66) Turning to FIG. 5, an exemplary environment **500** for validating and completing workflows and steps in a dynamic UX application is depicted, in accordance with some embodiments.

Environment **500** may include a client **502** which may correspond to mobile computing device **104** and/or client device **304**. Environment **500** may include a workflow module **504** which may correspond to workflow module **126**, workflow module **204**, and/or workflow module **308**.

Environment **500** may further include a system of interaction **520**, a business capability service **540** and a system of record **542**. System of interaction **520** may be integral to an experience manager, such as experience manager **206**, according to some embodiments. Workflow module **504** may include workflows W.sub.1-W.sub.m, wherein m is any integer, and which may respectively contain steps S.sub.1-S.sub.n, S.sub.1-S.sub.o, and S.sub.1-S.sub.p wherein n, o, and p may be any integers. That is, the steps in workflows W.sub.1-W.sub.m may respectively include any number of steps. The steps may be ordered. However, it should be noted that the number of steps per workflow may differ. Therefore, in the depicted example, the indices n, o, and p may differ, in addition to the identity of the steps composing each respective workflow. Further, no two steps at the same position within a workflow may necessarily be the same (e.g., W.sub.1S.sub.1 and W.sub.2S.sub.1 may differ). Workflow module **504** may include workflow step catalog **506** and impact analyzer **508**, which may correspond to step analyzer **212**. Workflow step catalog **506** may be an electronic database which may or may not be integral to workflow module **504**. For example, in some embodiments, workflow step catalog **506** may be a remote database communicatively coupled to a device in which workflow module **504** is included, via a network. Impact analyzer **508** may determine what information the workflow currently has, and what additional information may be required to complete the workflow. For example, impact analyzer **508** may determine available information (e.g., username, client ID, and level of assurance) and additional required information. Impact analyzer may determine which additional information is required by accessing system of interaction **520**.

(67) In an embodiment, workflow module **504** may load a workflow by a name provided in a uniform resource identifier (URI). The workflow may comprise a list of step references. Workflow module **504** may hydrate the workflow by instantiating an instance of each step reference in the workflow. Workflow module **504** may load step classes from step catalog **506**. As workflow module **504** instantiates each step reference, workflow module **504** may associate the instantiated class with the respective step reference in the workflow being hydrated. The hydrated workflow may then be passed to a runtime. Steps may be annotated with canonical resource and type information as they are instantiated.

(68) In some embodiments, the additional required information determined by impact analyzer **520** may be grouped by step. For example, in a credit card application workflow, the impact analyzer **508** may determine that the workflow includes the following steps and respective required information:

(69) TABLE-US-00001 Step Required Information Primary Applicant Basic Client Data Name, Home Address Primary Application Expanded Credit Birth Date, Social Card Client Data Security Number Primary Application Basic Income Gross Income State Activation Lookup State of Residence

(70) Impact analyzer **508** may cause workflow module **504** to modify the ordering of steps within the workflow, or to move the current step pointer to a particular step in response to missing information, or in response to a question in a step being answered in a specific way, or in response to a user-initiated event. For example, impact analyzer **508** may cause workflow module **504** to

redirect the user to a step that indicates that the workflow is unable to continue the steps because the step needs missing information from the user. Once the user provides the additional information that is required, by sending a response received by workflow manager **304**, impact analyzer **308** may analyze what steps can be executed with the new information.

(71) Impact analyzer **508** may query information index **528**. Information index **528** may be a data store that persists the information that is collected from the user or by system of interaction **520** on the user's behalf during an interaction. Similarly, impact analyzer **508** may transmit information to system of interaction **520** for storage and future use, including during the life of the interaction.

(72) In an embodiment, impact analyzer **508** may query aggregator **522** for relevant business capability information. Aggregator **522** may more granularly determine, based on the required information, what information is available from various data sources (e.g., both internal and external data APIs) and how that information may be obtained. For example, aggregator **522** may collect information required by workflow steps using information available in information index **528** as well as information available by calling, for example, business capability service **540** or external vendor services. To retrieve information from business capability service, aggregator **522** may use information contained within information index **528** in conjunction with service registry **530** to determine what services can be called to retrieve the required information.

(73) Aggregator **522** may gather information needed by workflow steps from service registry **530** based on information from information index **528** without requiring additional user input. Service registry **530** may use an algorithm (e.g., a best fit analysis algorithm) to select a service that can be called by the aggregator **522** that provides the most optimal access to the information. Under certain business circumstances, information should not be able to be received from aggregator **522** even if it is available. In that case, the information may be required to be retrieved from the customer, if at all. Some information may be discoverable in some contexts via aggregator **522**, but required to be elicited from the customer in others.

(74) Data pertaining to users may be collected by aggregator **522**, and a user may only be prompted to provide information that is required. For example, system of interaction **520** and/or aggregator **522** may determine that the following user information exists in the information index with respect to user Jane Smith:

(75) TABLE-US-00002 Name Residence Email Phone SSN Employer Income Jane 1600
jane@example.com 555-1234 123-45-6789 — — Smith Pennsylvania Ave.

(76) In this case, the missing information includes employer and income. For example, employer may be required, and income optional. Then, the user will be only prompted for employer, or may be prompted to supply both but a UI control may be accentuated to indicate to the user that income is not required information. In an embodiment, resource types may be included in information index **528**. Steps may be required to verify that information passed to them is of the correct type, and they can use resource types for that verification. Steps may also need to access interaction metadata. In an embodiment, workflow module **504** may include a workflow W. The workflow may include instructions which cause the workflow manager to look up and invoke services when workflow W is executed. In some embodiments, workflow W may specify only which data is needed, but not which services are to be used to obtain that data. Therefore, workflow creators may create workflows and reference business data elements without needing to have specific knowledge of where that data will ultimately be sourced from, or where that data needs to be stored.

(77) In general, workflows may be granted access to business context, and such context may be incorporated into the execution of steps within the workflow. For example, a business context may include the yearly fee of a credit card. As the steps in the workflow execute, they may make reference to the yearly fee via the business context. If the yearly fee changes, then the fee may be updated for the steps using the context.

(78) In operation, a user of client **502** may click on a link (e.g., a credit card application link) or activate another suitable user interface element. In response to the activation, a request may be

generated by way of a protocol handler. The protocol handler may implement a JSON API (e.g., Katharsis) that maps requests to resources (e.g., java classes). The resource(s) may call workflow module 504, and workflow module 504 may process the request and return a response. For example, java classes representing resources may be implemented as:

(79) TABLE-US-00003 Ccapp.java @JSONAPI Resource (type="ccpapps") data attributes of cc, address, name, etc class ccapp{ ccapp repository,java @JSONAPI resource repository
(CCApp.class class ccApp findAll finddone saveC delete

(80) The functions in the resources may invoke workflow module 504 by passing a request to the workflow module 504. Workflow module 504 may proceed to execute steps as described above. Workflow module 504 may access impact analyzer to determine what information workflow module 504 has and what is needed to complete workflow steps. For example, referring to the table above, workflow module 504 may determine that it is unable to complete the steps within a workflow because the information the workflow module 504 has is limited to username, client ID, and LOA3. Then, workflow module 504 may access aggregator 522 if workflow module 504 determines that it cannot continue with the information workflow module 504 has, and may provide aggregator 522 with the information workflow module 504 currently has. For example, to continue the above example, workflow module 504, having determined that it cannot proceed because it lacks needed information, may access aggregator 522, providing username, client ID, and LOA3 as parameters.

(81) Aggregator 522 may then use known information to query for additional information. Aggregator 522 may query information index 528 for stored information. Available information (if any) may be passed from information index 528 to aggregator 522. Aggregator 522 may query business capability service 540 for relevant services. If relevant services are found, then a list of the available services may be returned to aggregator 522. If aggregator 522 receives an indication of relevant services from business capability service 540, then aggregator 522 may call each of those services, in parallel or in serial, and subject to any access control limits placed on aggregator 522 with respect to those services. In addition to aggregator 522, steps in a workflow and system of action 520 may invoke business capability service 540 or other services/process APIs. Information from system of interaction 520 may be transformed from canonical message format to a format specific to business capability service 540 before being sent to business capability service 540. Similarly, results from business capability service 540 may be translated into canonical message format before being transferred by system of interaction 520. Translation may involve the creation of a dynamic web service and API call, wherein the API call includes input properties.

(82) Once workflow module 504 has accessed aggregator 522 and retrieved information obtained from relevant services, workflow module 504 may input the information into the executed steps. In particular, missing information (e.g., employer and income) may be added and the step may be considered complete. If the step is considered complete, then the workflow module 504 may proceed to execute the next step according to a step ordering or other algorithm (e.g., a tree traversal algorithm, in an embodiment wherein the steps are hierarchically ordered). In some embodiments, an indication of step completion may be transmitted to client 502. In some embodiments, workflow module 504 may be unable to determine missing information. In such cases, workflow module 504 may include instructions which cause a response to be generated which includes, as described above, a dynamic UX application (or portion thereof) soliciting the missing information from the user. The response may be transmitted to client 502 for display and collection.

IV. Additional Example Environments for Generating Dynamic UX Applications

(83) An experience manager may be a presentation layer component that is embedded with client. In that case, there may be a clear separation of concerns between presentation and business layers, but any change in the experience manager may require time-consuming UI application build and deployment with additional regression testing. Reconfiguring the UI based on a user profile or

activity may be very difficult. Further, changes to the canonical data model may require rebuilding and redeploying any UI components. In general, the purpose of the experience manager component is to return UI controls dynamically based on individualized UX rules. It should be appreciated that there are many ways to accomplish this goal, and the optimal abstractions may depend on the underlying hardware configuration. For example, a thin client or underpowered mobile device may have different device capabilities than a top-of-the-line smart phone, and may demand alternate methods of delivering dynamic UX applications. By separating workflow/step execution, and user experience modules (and related components) and building consistent APIs in front of them, modifying or reconfiguring user interfaces and user experience may be done very quickly without affecting client code or step/workflow execution.

(84) Turning to FIG. 6, an example environment **600** for generating workflow-based dynamic UX applications which separate the application components as a series of APIs or services is depicted, according to an embodiment. Environment **600** may include a client **602** which may correspond to client **502**, user device **304**, and client **104**. Environment **600** may further include an application **604**, which may correspond to application **122** and/or application **168**. Environment **600** may further include an experience manager API **606** and business API **608**. Experience manager API may correspond to, or implement, experience module **310**, experience manager **206**, and experience module **132**.

(85) The embodiment depicted in environment **600** may help to separate the presentation logic-related components from the business logic layer. Modules may be organized according to function, and may be wrapped in an API layer which allows the modules to be accessed indirectly. In an embodiment, application **604** may call business API **604** first and experience manager **612** second, after receiving a response from business API **408**. In this way, experience manager **612** may be an optional component for the dynamic UX experience consumer. Clear separation of concerns between presentation and business layers may be achieved and presentation and business layers may operate completely independently of one another. Application **604** may be able to issue API calls to business API **608**, and need only be programmed to access a uniform business API. Then, various components of business API may be mapped to those calls (e.g., workflow module **620**). Additionally, business API **608** may include instructions to access experience manager API **606**, which may determine UX. In some embodiments, some or all of dynamic UX applications may be cached in content delivery network (CDN) **610a**, which may be accessed by application **604**. Also, UX may be determined by a configurable service such as experience manager **606** using a UX rules component **616** accessing a UX rules database **610b**. Experience manager **612** may also determine user experience by querying system of interaction behavior and events **612**, which may contain historical user preferences and events related to past user behavior. Regardless of whether UX is being determined based on user experience rules **616** or system of interaction behavior and events **612**, business API will make the same API call. Therefore, workflow module **620** need not include specific logic for acquiring a user experience-workflow module **620** need only include instructions for querying experience manager API **606**.

(86) Dynamic UX application consumers (e.g., business API **608**) may access experience manager API **606** based on need. Page layout, input validation (meta data), and page styling can be quickly changed from the configurable user experience manager **606**. Any change or additions of new UI fields (e.g., adding a new question) may be externalized and configured in experience manager **606**, and no change to the client **602** is necessary if the user experience changes.

(87) With regard to FIG. 7, a data flow diagram for multiple workflow execution is depicted, according to an embodiment. As shown, a user may submit a request from a client **702**, which may correspond to client **104**, client device **304**, client **502**, and/or client **602**. The request may be received a protocol handler **704**, which may correspond to protocol handler **136**. The request may be translated/processed by protocol handler **704**, and passed to workflow module **710**. Workflow module **710** may correspond to business API **608**, workflow module **504**, workflow module **308**,

workflow module **204**, and/or workflow module **126**. In general, workflow module **710** may be located in a server, but may also be integral to a client. Workflow module may examine the request from client **702**, and may begin executing a workflow. Execution of the workflow may be performed by workflow state machine **720**. Workflow module **710** may transmit output of the executing workflow in whole or in part to experience module **706**, where dynamic UX may be applied to the output by experience module **706**. For example, experience manager **706** may comprise an experience engine that performs a number of functions. First, experience manager **706** may retrieve customer (e.g., applicant) information from information index **716** for display and UX rule execution. Experience manager **706** may retrieve user behaviors and/or UX events for UX rule execution from behaviors and UX events **718** within system of interaction **708**. An interaction ID may be passed as a parameter to the system of interaction **708** to look up information corresponding to the current interactions associated with the customer. Experience engine **706** may execute UX rules to determine how to collect/present information in client **702**. For example, the sequencing of UI elements, controls to use, etc. may be determined by the execution of UX rules. Experience engine **706** may also determine a template for the overall layout, which may be provided as executable code to client **702**. Experience module **706** may prepare canonical data for transmission to protocol handler **704**, and may prepare metadata for each canonical data element.

(88) Addition of dynamic UX may be based on analyzing behaviors and UX events **718**. Behaviors and UX events **718** may include actions that a user of client **700** has taken when using the environment **700** previously (e.g., the speed at which the user has proceeded through steps in a workflow). The user may be associated with client **700** and behaviors and UX events **718**, respectively, by a session, cookie, user identifier, or other suitable data. After dynamic UX is added, in canonical data format, the data may be passed to protocol handler **704**, which may convert the canonical data format to a format that can be read by client **702**. As shown, a REST API **714** may be used to translate client requests and responses to and from canonical data format. As depicted, client **700** may include a UI control library which implements various UI components that may be used in the rendering of the dynamic UX application and portions thereof. UI control **712** identifiers may map controls in client **700** to UI fields (e.g., text field, drop-downs, etc.). Experience manager **706** may generate a relative sequence for display elements, mark fields as editable or read-only, and/or inject scripts into a dynamic UX application. For example, a hide/show indicator may be inserted by experience manager **706**.

(89) The dynamic UX application may be generated by workflow module **710**. Specifically, in operation, a request may be received by workflow module **710** in which a customer seeks to apply for an insurance quote. The request may pass through protocol handler **704** and may include an interaction ID. Workflow manager **710** may cause workflow state **720** to begin executing a first workflow **724**, which may execute a series of steps. Workflow state **720** may then initiate a second workflow **726**, which may include steps **732a-732c**. Before executing step **732a**, workflow **726** may request applicant information from system of interaction **708**, passing interaction ID as a parameter. System of interaction **708** may locate an interaction having the customer/applicant's name, gender, birth date, and phone number. System of interaction **708** may provide an indication of the information it has to workflow manager **710**. Next, workflow state **720** may determine that driver's license number is missing from the information provided by system of interaction **708**. Workflow manager **710** may then send a response to experience manager **706** indicating that metadata is missing, specifically, the customer's driver's license number. Experience manager **706** may then craft a response prompting the customer for his or her driver's license number, based on the customer's user preferences. Experience manager **706** may pass that response to protocol handler **704**, which may translate the response and transmit it via REST API **714** to client **702**, where it may be displayed by UI control library **712** to the user. The user may provide the driver's license number, at which point the workflow may proceed.

(90) Experience manager **706** may have access to pieces of information that enable it to better

personalize the UX. For example, experience manager **706** may access a user's identity or persona/segment, current business context/objective, previous interactions via system of interaction **708**, and business processes' information of the customer. Many pieces of information may be analyzed in personalizing a user's experience. The examples herein discuss details pertaining to the user's use patterns and/or personal attributes, however, in some embodiments other external information (e.g., time, location, and weather) may be factored into the UX by the experience manager **706**.

(91) Once step **726** is complete, workflow state **720** may execute workflow **728**. Workflow **728** may include a single step, which may transmit the user's application, and may also contain logic for processing a result of the submission. For example, the submission may be approved, denied, or pended. The step may notify a user or take another action depending upon the result of the submission.

ADDITIONAL CONSIDERATIONS

(92) With the foregoing, any users (e.g., insurance customers) whose data is being collected and/or utilized may first opt-in to a rewards, insurance discount, or other type of program. After the user provides their affirmative consent, data may be collected from the user's device (e.g., mobile device or other smart devices). Of course, local storage and use of a disposable UI at a user device (e.g., the client device **102** of FIG. **1**) by an anonymous user may have the benefit of removing any concerns of privacy or anonymity, by removing anything except the minimum user interface elements needed to render steps and workflows, without storing data permanently on the client device. In such instances, there may be no need for affirmative consent.

(93) Although the text herein sets forth a detailed description of numerous different embodiments, it should be understood that the legal scope of the invention is defined by the words of the claims set forth at the end of this patent. The detailed description is to be construed as exemplary only and does not describe every possible embodiment, as describing every possible embodiment would be impractical, if not impossible. One could implement numerous alternate embodiments, using either current technology or technology developed after the filing date of this patent, which would still fall within the scope of the claims.

(94) It should also be understood that, unless a term is expressly defined in this patent using the sentence "As used herein, the term '_____' is hereby defined to mean . . ." or a similar sentence, there is no intent to limit the meaning of that term, either expressly or by implication, beyond its plain or ordinary meaning, and such term should not be interpreted to be limited in scope based upon any statement made in any section of this patent (other than the language of the claims). To the extent that any term recited in the claims at the end of this disclosure is referred to in this disclosure in a manner consistent with a single meaning, that is done for sake of clarity only so as to not confuse the reader, and it is not intended that such claim term be limited, by implication or otherwise, to that single meaning. Unless a claim element is defined by reciting the word "means" and a function without the recital of any structure, it is not intended that the scope of any claim element be interpreted based upon the application of 35 U.S.C. § 112(f). The systems and methods described herein are directed to an improvement to computer functionality, and improve the functioning of conventional computers.

(95) Throughout this specification, plural instances may implement components, operations, or structures described as a single instance. Although individual operations of one or more methods are illustrated and described as separate operations, one or more of the individual operations may be performed concurrently, and nothing requires that the operations be performed in the order illustrated. Structures and functionality presented as separate components in example configurations may be implemented as a combined structure or component. Similarly, structures and functionality presented as a single component may be implemented as separate components. These and other variations, modifications, additions, and improvements fall within the scope of the subject matter herein.

(96) Additionally, certain embodiments are described herein as including logic or a number of routines, subroutines, applications, or instructions. These may constitute either software (code embodied on a non-transitory, tangible machine-readable medium) or hardware. In hardware, the routines, etc., are tangible units capable of performing certain operations and may be configured or arranged in a certain manner. In example embodiments, one or more computer systems (e.g., a standalone, client or server computer system) or one or more modules of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a module that operates to perform certain operations as described herein.

(97) In various embodiments, a module may be implemented mechanically or electronically. Accordingly, the term “module” should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering embodiments in which modules are temporarily configured (e.g., programmed), each of the modules need not be configured or instantiated at any one instance in time. For example, where the modules comprise a general-purpose processor configured using software, the general-purpose processor may be configured as respective different modules at different times. Software may accordingly configure a processor, for example, to constitute a particular module at one instance of time and to constitute a different module at a different instance of time.

(98) Modules can provide information to, and receive information from, other modules. Accordingly, the described modules may be regarded as being communicatively coupled. Where multiple of such modules exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) that connect the modules. In embodiments in which multiple modules are configured or instantiated at different times, communications between such modules may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple modules have access. For example, one module may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further module may then, at a later time, access the memory device to retrieve and process the stored output. Modules may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information).

(99) The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented modules that operate to perform one or more operations or functions. The modules referred to herein may, in some example embodiments, comprise processor-implemented modules.

(100) Similarly, the methods or routines described herein may be at least partially processor-implemented. For example, at least some of the operations of a method may be performed by one or more processors or processor-implemented modules. The performance of certain of the operations may be distributed among the one or more processors, not only residing within a single machine, but deployed across a number of machines. In some example embodiments, the processor or processors may be located in a single location (e.g., within a home environment, an office environment or as a server farm), while in other embodiments the processors may be distributed across a number of locations.

(101) The performance of certain of the operations may be distributed among the one or more processors, not only residing within a single machine, but deployed across a number of machines. In some example embodiments, the one or more processors or processor-implemented modules may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other example embodiments, the one or more processors or

processor-implemented modules may be distributed across a number of geographic locations. (102) Unless specifically stated otherwise, discussions herein using words such as “processing,” “computing,” “calculating,” “determining,” “presenting,” “displaying,” or the like may refer to actions or processes of a machine (e.g., a computer) that manipulates or transforms data represented as physical (e.g., electronic, magnetic, or optical) quantities within one or more memories (e.g., volatile memory, non-volatile memory, or a combination thereof), registers, or other machine components that receive, store, transmit, or display information. Some embodiments may be described using the expression “coupled” and “connected” along with their derivatives. For example, some embodiments may be described using the term “coupled” to indicate that two or more elements are in direct physical or electrical contact. The term “coupled,” however, may also mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other. The embodiments are not limited in this context.

(103) As used herein any reference to “one embodiment” or “an embodiment” means that a particular element, feature, structure, or characteristic described in connection with the embodiment may be included in at least one embodiment. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment. In addition, use of the “a” or “an” are employed to describe elements and components of the embodiments herein. This is done merely for convenience and to give a general sense of the description. This description, and the claims that follow, should be read to include one or at least one and the singular also includes the plural unless it is obvious that it is meant otherwise.

(104) As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

(105) This detailed description is to be construed as exemplary only and does not describe every possible embodiment, as describing every possible embodiment would be impractical, if not impossible. One could implement numerous alternate embodiments, using either current technology or technology developed after the filing date of this application. Upon reading this disclosure, those of skill in the art will appreciate still additional alternative structural and functional designs for generating dynamic user experience applications through the disclosed principles herein. Thus, while particular embodiments and applications have been illustrated and described, it is to be understood that the disclosed embodiments are not limited to the precise construction and components disclosed herein. Various modifications, changes and variations, which will be apparent to those skilled in the art, may be made in the arrangement, operation and details of the method and apparatus disclosed herein without departing from the spirit and scope defined in the appended claims.

(106) The particular features, structures, or characteristics of any specific embodiment may be combined in any suitable manner and in any suitable combination with one or more other embodiments, including the use of selected features without corresponding use of other features. In addition, many modifications may be made to adapt a particular application, situation or material to the essential scope and spirit of the present invention. It is to be understood that other variations and modifications of the embodiments of the present invention described and illustrated herein are possible in light of the teachings herein and are to be considered part of the spirit and scope of the present invention.

(107) While the preferred embodiments of the invention have been described, it should be understood that the invention is not so limited and modifications may be made without departing

from the invention. The scope of the invention is defined by the appended claims, and all devices that come within the meaning of the claims, either literally or by equivalence, are intended to be embraced therein. It is therefore intended that the foregoing detailed description be regarded as illustrative rather than limiting, and that it be understood that it is the following claims, including all equivalents, that are intended to define the spirit and scope of this invention.

Claims

1. A computer-implemented method, comprising: identifying, by a processor, a workflow including a set of one or more computer-executable steps that are incomplete; transmitting, by the processor and to an electronic device, an executable application which, when executed by the electronic device, causes the electronic device to present a first user interface via a display, the first user interface being operable to receive one or more inputs required to complete a current step of the set of one or more computer-executable steps that are incomplete; receiving, by the processor, an input entered via the first user interface executed by the electronic device; determining, by the processor, that the input includes information required to complete the current step; completing, by the processor, the current step based on the information; based on completing the current step, and in real time: determining, by the processor, a next step of the set of one or more computer-executable steps that are incomplete; and dynamically modifying, by the processor, the executable application such that, when executed by the electronic device, the modified application causes the electronic device to present a second user interface via the display, the second user interface indicating information required to complete the next step; and transmitting, by the processor, the modified application to the electronic device, the modified application causing the electronic device to present the second user interface via the display.
2. The method of claim 1, further comprising: determining, by the processor and based on completing the current step, a sequence for executing the set of the one or more computer-executable steps, wherein the next step follows the current step in the sequence.
3. The method of claim 2, wherein the processor determines the sequence by applying a machine learning algorithm to the set of one or more computer-executable steps, the machine learning algorithm being configured to output the next step based on completing the current step.
4. The method of claim 2, wherein the input is a first input, the method further comprising: receiving, by the processor, a second input entered via the second user interface; and determining, by the processor and based on the second input, a reordering of the sequence for executing the set of the one or more computer-executable steps.
5. The method of claim 1, wherein the application is based on at least one of: a capability of the electronic device, a communication protocol of the electronic device, or profile information associated with a user of the electronic device.
6. The method of claim 1, further comprising: receiving, by the processor and from the electronic device, a request; and determining, by the processor, a match between a search term in the request and one or more terms associated with the workflow, wherein identifying the workflow is based at least in part on determining the match.
7. The method of claim 1, further comprising: accessing, by the processor, a user profile associated with the electronic device; and determining, by the processor and based at least in part on the user profile, a permission associated with the user profile, wherein the workflow is identified based at least in part on the permission.
8. The method of claim 7, wherein the user profile indicates a role of a user associated with the electronic device, the user profile comprising at least one of: an insurance agent, an insured customer, or a customer service representative.
9. The method of claim 1, wherein the input is a first input, the method further comprising: receiving, by the processor, a second input from an information source different from the electronic

device, wherein the second user interface provides the second input and an indication of the information source.

10. The method of claim 1, wherein each step of the set of one or more computer-executable steps is associated with a respective user interface included in a list of user interfaces, the method further comprising: selecting, by the processor, the second user interface from the list of user interfaces, based on the second user interface being associated with the next step.

11. The method of claim 2, wherein an order of execution of the sequence is based at least in part on a level of confidentiality associated with information required to execute a computer-executable step included in the set of the one or more computer-executable steps.

12. A system, comprising: a processor; and non-transitory computer-readable memory storing computer-executable instructions that, when executed by the processor, cause the processor to perform operations comprising: identifying a workflow including a set of one or more computer-executable steps that are incomplete; transmitting, to an electronic device, an executable application which, when executed by the electronic device, causes the electronic device to present a first user interface via a display, the first user interface being operable to receive one or more inputs required to complete a current step of the set of one or more computer-executable steps that are incomplete; receiving an input entered via the first user interface executed by the electronic device; determining that the input includes information required to complete the current step; completing the current step based on the information; based on completing the current step, and in real time: determining a next step of the set of one or more computer-executable steps that are incomplete; and dynamically modifying the executable application such that, when executed by the electronic device, the modified application causes the electronic device to present a second user interface via the display, the second user interface indicating information required to complete the next step; and transmitting the modified application to the electronic device, the modified application causing the electronic device to present the second user interface via the display.

13. The system of claim 12, the operations further comprising: accessing a user profile associated with the electronic device, wherein the user profile indicates a role of a user of the electronic device; and determining characteristics of the electronic device, wherein the first user interface and the second user interface are based at least in part on the role of the user and the characteristics of the electronic device.

14. The system of claim 12, wherein the electronic device is a mobile device, and the application is a dynamic user experience application configured to execute as a standalone application on the mobile device.

15. The system of claim 12, wherein the input is a first input, the operations further comprising: receiving a second input from an information source, wherein the second user interface provides an indication of the information source.

16. The system of claim 12, the operations further comprising: determining, based on a level of confidentiality associated with information required to execute a computer-executable step, a sequence for executing the set of the one or more computer-executable steps, wherein the next step is determined as sequentially following the current step in the sequence.

17. A tangible, non-transitory computer-readable medium storing instructions that, when executed by a processor, causes the processor to: identify a workflow including a set of one or more computer-executable steps that are incomplete; transmit, to an electronic device, an executable application which, when executed by the electronic device, causes the electronic device to present a first user interface via a display, wherein the first user interface is operable to receive one or more inputs required to complete a current step of the set of one or more computer-executable steps that are incomplete; receive an input entered via the first user interface executed by the electronic device; determine that the input includes information required to complete the current step; complete the current step based on the information; based on completing the current step, and in real time: determine a next step of the set of one or more computer-executable steps that are

incomplete; and dynamically modify the executable application such that, when executed by the electronic device, the modified application causes the electronic device to present a second user interface via the display, the second user interface indicating information required to complete the next step; and transmit the modified application to the electronic device, the modified application causing the electronic device to present the second user interface via the display.

18. The tangible, non-transitory computer-readable medium of claim 17, the instructions further causing the processor to: determine a sequence for executing the set of one or more computer-executable steps, wherein the next step follows the current step in the sequence; and determine, based on the input, a reordering of the sequence for executing the set of the one or more computer-executable steps.

19. The tangible, non-transitory computer-readable medium of claim 17, wherein: the electronic device is a mobile device associated with one or more characteristics, and the first user interface and the second user interface are selected based on the one or more characteristics.

20. The tangible, non-transitory computer-readable medium of claim 17, the instructions further causing the processor to: receive, from the electronic device, a request; and determine a match between a search term in the request and one or more terms associated with the workflow, wherein identifying the workflow is based at least in part on determining the match.
